

SEMESTER END EXAMINATIONS – FEB / MARCH 2025

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Computer Networks	Max. Marks	: 100
Course Code	: IS51(000) (Prior to 2021 Batch)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- What is the purpose of a subnet mask in IPv4 addressing? What are the challenges of migrating from IPv4 to IPv6, and what steps are involved in such a migration? CO1 (10)
 - What is the difference between static NAT and dynamic NAT? CO1 (06)
 - What are the IPv4 address classes, and how do they differ (Class A, B, C, D, E)? CO1 (04)
- What are the key differences in the address notation between IPv4 and IPv6? CO1 (06)
 - Explain the concept of NAT mapping and how NAT devices maintain a translation table for connections. CO1 (06)
 - How does subnetting allow the division of an IPv4 address space into smaller subnets? Provide an example. CO1 (08)

UNIT - II

- Discuss the formation of least cost trees using Dijkstra's Algorithm. CO2 (08)
 - Differentiate between Virtual Circuit approach and Datagram Approach. CO2 (08)
 - Discuss the network layer services. CO2 (04)
- Differentiate between link state routing and path vector routing with an example for each. CO2 (08)
 - Describe the following network layer performance CO2 (06)
 - Delay
 - Throughput.
 - Discuss the following: CO2 (06)
 - DHCP
 - ARP
 - RARP.

UNIT - III

- Explain the three-way handshake in TCP connection establishment. How does it ensure reliable communication with the help of suitable diagram. CO3 (10)
 - What is a socket, and how does it relate to Transport Layer protocols like TCP and UDP? CO3 (04)
 - How does SCTP differ from TCP and UDP in terms of its functionality and features? CO3 (06)

6. a) How does SCTP manage congestion control, and how does it compare to TCP in terms of congestion control mechanisms? CO3 (08)
b) Explain the concept of a "port" in the Transport Layer and its importance in identifying applications. CO3 (04)
c) Why is UDP chosen over TCP in applications like live video streaming, and what measures are taken to handle issues like packet loss? What is "flow control" in TCP, and how does it ensure that a sender does not overwhelm the receiver with data? CO3 (08)

UNIT- IV

7. a) What role does SMTP play in sending email? What are the main components involved in an SMTP transaction? CO4 (08)
b) Explain the difference between a domain name and an IP address. CO4 (04)
c) What is the role of SNMP in network performance analysis and fault detection? CO4 (08)
8. a) Illustrate SMTP with respect to : CO4 (08)
i) Message transfer agent
ii) Commands and responses
iii) Header Translation.
b) How does DNS resolution differ in IPv4 and IPv6? CO4 (06)
c) How is SNMP used to monitor network devices like routers and switches? CO4 (06)

UNIT - V

9. a) How does a digital signature provide authentication and integrity for a message or document? CO5 (08)
b) Explain the Working Principle of Diffie-Hellman key exchange algorithm with a suitable example. CO5 (08)
c) What role do large prime numbers play in the RSA algorithm? CO5 (04)
10. a) Illustrate the following with the help of the neat diagram: Model for CO5 (08)
i) Network Security
ii) Simplified model for symmetric Encryption.
b) Perform encryption and decryption using the RSA algorithm for the CO5 (08)
following:
i) $P=3, q=13, e=5, M=10$
ii) $P=5, q=7, e=7, M=12$
c) Explain the mathematical process behind RSA digital signatures, CO5 (04)
including how the signing and verification operations work.
